

UNIT-5: INTELLECTUAL PROPERTY MANAGEMENT

Creating Intellectual Property: Need for creating intellectual property – Development of IP and Knowledge – Types of innovations – Behavioral aspects. Intellectual Property Management: Need and importance of IP management – IP management activities – 5Cs model of managing IP – Research and Developments in India (Case Study).

NEED FOR CREATING INTELLECTUAL PROPERTY

Intellectual property rights not only protect the ideas or concepts of business but also protect the genuine business assets that are vital to the products and services.

The need for creation of intellectual property is envisaged due to following reasons:

- **Wealth creation**
 - **Identity in the marketplace**
 - **Entering in new markets**
 - **Creation of entry barriers**
 - **Strengthening and safeguarding the existing IP**
 - **Shaping industry architecture**
 - **Protecting traditional knowledge**
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- **Wealth creation:** Intellectual property rights can help you generate business through the licensing, sale and even commercialization of the products and services protected under IPRs. This will ultimately improve the market share and helps in raising profits. Having registered and protected intellectual property rights can also raise the business' wealth
 - **Identity in the market place:** Getting intellectual property rights can help the business' image. Intellectual property rights like trademark registration can help to separate the products and services from others
 - **Entering new markets:** A business that has registered IPRs will be able to use brands and designs to market its products and services to other markets as well. A business can also tap into the franchising agreements with overseas companies or export patented products.
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1. **Creation of entry barriers:** A barrier to entry is any obstacle that limits or prevents others from competing directly with company's product or service. Registered intellectual property like patent and trade secrets serve as a potential barrier to entry. A strong and well-drafted utility patent gives the owner the right to prevent any other party from making, using or selling the patented invention in the territory covered by the patent. Potential competitors face risk.
 2. **Strengthening and safeguarding existing IP:** A proactive approach to intellectual property protection is not only a legal necessity but a strategic imperative for the sustained growth of startups in India. By leveraging legal frameworks, adopting best practices, and utilizing available government resources, startups can ensure their innovations are shielded, paving the way for long-term success in the competitive business landscape.

3. **Shaping industry architecture:** Intellectual property comes in many forms. They can be in the form of books, music, designs, processes, marks, geographical indications or even data or information's used in industry or commerce. There is a trend that the scope and form of intellectual property will be further expanded in the future. Most objects involve in day-to-day life such as furniture's, stationary, kitchenette, automobiles, computers, telecommunication equipment's, etc. did not exist thousands of years ago. They are not natural objects but are made by man, the fruits of intellectual property.
4. **Protecting traditional knowledge:** is incredibly essential to preserve traditional knowledge, especially in these ever-changing times. Protecting traditional knowledge can stop unauthorized, commercial misuse of this knowledge base. If it remains unprotected, indigenous people who are responsible for bringing it to the forefront can end up suffering huge losses, on an emotional as well as a financial level. By protecting traditional knowledge, one can protect and preserve ancient practices.

DEVELOPMENT OF IP AND KNOWLEDGE

A variety of input factors, including social, cultural, economic, genetic, and legal influences, frame individual understanding around intellectual property rights. This psychological understanding of intellectual property rights, in turn, drives behavioral activity concerning individual decisions to engage in creative endeavors, invest resources in innovation, and to observe (or not) the intellectual property rights of others.

- Human Thought Process
 - Tacit Ideas/Knowledge
 - Codification
 - Validation/Experimentation
 - Explicit Scientific/Business Knowledge
 - Scrutiny
 - Intellectual Property
- **Human thought process:** The success of intellectual property (IP) systems depends on their ability to influence human behavior in relation to creativity and innovation. Intellectual property (IP) pertains to any original creation of the human intellect such as artistic, literary, technical, or scientific creation. Intellectual property rights (IPR) refer to the legal rights given to the inventor or creator to protect his invention or creation for a certain period of time
 - **Tacit ideas and knowledge:** Tacit knowledge is a valuable input into innovation processes. Knowledge obtained by experience or by interaction can differ in impact. Awareness, critical thinking, and willingness to share enhance tacit effectiveness. Innovation type (product or process) varies in how tacit influences outcomes.

- **Codification:** It is the act, process, or result of stating the rules and principles applicable in a given legal order to one or more broad areas of life in this form of a code
- **Validation/experimentation:** Product validation is the process of testing an idea with potential users to get feedback on a product's viability. It checks whether a product or feature addresses the needs and pain points of current and potential customers. **Engaging in product validation early in the development process strategically direct time and funds to create the best product possible. Experimentation is a form of product validation.** It's most useful to assess iterations and improvements to a feature or product once it's already been built. Validation can help inform the direction of product experiments and help to reduce overall assumptions.
- **Explicit scientific / business knowledge:** Discoveries made through scientific research can have great value—to researchers in advancing knowledge, to governments in setting public policy, and to industry in developing new products.
- **Scrutiny:** The value of an IP asset essentially comes from the right the owner of that asset has to exclude competitors from using it. For an IP asset to have a quantifiable value it should: generate a measurable number of economic benefits to its owner/user; and. enhances the value of other assets with which it is associated.
- Intellectual property is crucial that enterprises identify and develop their knowledge base, not least in their market relationships. Knowledge emerges in network relationships between an enterprise and its surrounding organizations, and can be legally protected as intellectual property (IP). A clearer notion of the way IP can be analyzed may improve the economic outcome of investments in innovation. Relationship marketing clearly has a role in the marketing of technological innovation and product development

INNOVATION

- Innovation can refer to **something new or to a change made to an existing product, idea, or field**

Types of innovations

Innovation drives progress and propels societies and businesses forward by introducing new ideas, products and processes.

There are four distinct types of innovation. These classifications help us understand the diverse ways innovation manifests itself and its impact on industries.

Types of innovation

Below you will find a general explanation of the four types of innovation as described by Henderson and Clark.

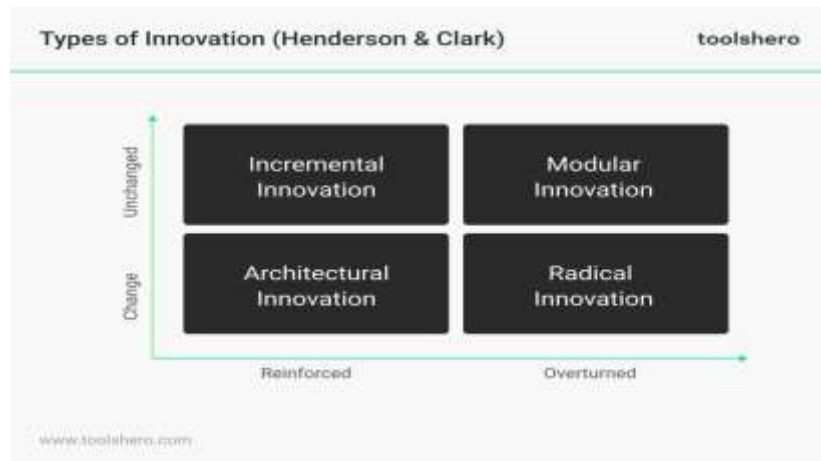


Figure 1 – 4 types of innovation (Henderson and Clark)

Type 1 – Incremental innovation

Incremental innovation refers to small, gradual improvements made to existing products, processes or services. It involves making incremental changes or improvements to improve efficiency, quality or functionality.

Example: The continuous updates and improvements made to popular software applications such as Microsoft Office, with each new version introducing new features, improved user experience, and better performance.

Benefits of incremental innovation:

- Enables continuous improvement of existing products, processes or services
- Minimizes risks, compared to radical changes, making it easier to implement and adopt
- Increases customer satisfaction by addressing incremental needs and preferences

Type 2 – Modular innovation

Modular innovation means developing independent components or modules that can be easily combined or exchanged to create different configurations. It offers flexibility, efficiency and customization options in design and production.

Example: Lego building blocks offer modular innovation where individual blocks can be combined and interchanged to create countless configurations and structures, allowing children to exercise their creativity and imagination.

Benefits of modular innovation:

- Facilitates faster development and implementation of new products or services
- Promotes scalability and customization through modular components
- Strengthens collaboration and innovation by allowing various organizations to contribute to a shared ecosystem

Type 3 – Architectural innovation

Architectural innovation involves reconfiguring the overall design or structure of a product, system, or organization. It involves making significant changes to the underlying framework or architecture to achieve new functionality, performance improvements, or cost efficiency.

Example: Tesla's electric cars are an example of architectural innovation by integrating advanced battery technology, electric drive systems and software systems.

This integration is revolutionizing the automotive industry by offering sustainable, high-performance vehicles with unique features such as over-the-air updates and autonomous driving.

Benefits architectural innovation:

- Enables the creation of new business models and value propositions
- Improves flexibility and adaptability by uncoupling components within a system
- Drives collaboration and partnership opportunities using modular interfaces

Type 4 – Radical innovation

Radical innovation involves introducing completely new products, processes or business models that significantly disrupt existing markets or create entirely new ones. It is often accompanied by groundbreaking and transformative developments.

Example: The introduction of the Internet and the World Wide Web has revolutionized communication, access to information and the way we do business. This radical innovation has brought about profound changes in several industries, enabling global connectivity, e-commerce and the digital transformation of countless aspects of our lives.

Benefits of radical innovation:

- Holds the potential for disruptive and transformative breakthroughs
- Creates new markets or radically reforms existing markets
- Offers competitive advantage and differentiation by introducing new and unique solutions

Summary on the 4 types of innovation

The Henderson and Clark framework is based on extensive research into innovation practices across industries.

The four types of innovation are:

1. Incremental innovation: Small, gradual improvements to existing products, processes, or services. Example: Ongoing updates in software applications such as Microsoft Office.
2. Modular innovation: Development of independent components that can be easily combined or exchanged. Example: Lego building blocks.
3. Architectural innovation: Reconfiguring the design or structure of a product, system, or organization. Example: Tesla's electric cars with advanced battery technology and software.
4. Radical innovation: The introduction of completely new products, processes or business models that disrupt existing markets or create new ones. Example: the Internet and the World Wide Web.

Behavioral aspects:

Behavioral aspects hinder IP management and administration. Hence the following aspects has to be followed to improve the number of intellectual properties.

- Open Atmosphere
- Tolerance for Failure
- Free Communication
- Quest for Knowledge
- Decentralization
- Management Support

- **Open Atmosphere:** The individuals and employees should be provided with an environment where any idea can be easily shared and transformed. The discouragement of new ideas and innovative thoughts should be completely avoided. All information and resources also should be available for the inventors and innovators.
- **Tolerance for Failure:** With the emphasis properly placed on the positive aspects arising from a failure, the failure is positioned as a great learning experience for all of those involved. By encouraging an open discussion of why things went wrong much can be gained and made available to the corporation. Managements who are intolerant of failure will ultimately kill initiative.
- **Free communication:** One of the easiest, most effective ways to encourage idea sharing is to make it clear that sharing ideas is acceptable. Think about it like this: People naturally shy away from embarrassment or negative feedback, so they might not share ideas unless given explicit permission. This can result in solely your most outspoken team members getting a say.
- **Quest for knowledge:** Knowledge is a powerful force that can open many doors of opportunity. It can provide us with the skills and understanding to tackle challenging tasks and improve our lives. Knowledge can also help us to gain insight into the world we live in, and equip us with the tools to make informed decisions and better navigate the intellectual property
- **Decentralization:** To maintain a competitive edge in today's economic environment, organizations must adjust quickly, demonstrating **flexibility and agility**, all of which can be handled through decentralization. Decentralization is a type of organizational structure that promotes more intensive **bottom-up communication** as well as a constant flow of ideas, information, and decisions. Slow advances in company performance can be attributed to a multi-layered organizational structure with decision-makers concentrated at the top-level management.
- In reality, decentralization entails distributing decision-making authority to lower-level management and employees. When employees are given this level of freedom and are rewarded for risk-taking, they become more creative in their pursuit of efficient solutions to satisfy unmet demands. Involving more people in decision-making, on the other hand, relieves pressure on top management, allowing them to focus on central issues.

- **Management Support:** Management support for innovation is not a one-time thing. It requires continuing top-level support. It is impossible in two months to create a complete plan when doing anything significant. If management doesn't support those efforts over time, no matter how good the idea, then there is no hope of success.

Intellectual Property Management

Intellectual Property management involves various strategies and practices to handle different types of intellectual property assets. Here are the different kinds of IP management:

1. **Patent Management** involves the strategic planning, filing, and maintenance of patents. It includes identifying patentable inventions, conducting patent searches, drafting patent applications, and managing a portfolio of granted patents.
2. **Trademark Management:** Managing trademarks involves selecting strong and distinctive marks, registering them with relevant authorities, monitoring potential infringement, and enforcing trademark rights to protect brand identity.
3. **Copyright Management:** Copyright management protects original creative works such as literature, music, software, and art. It includes registering copyrights, licensing works, and enforcing rights against unauthorized use.
4. **Trade Secret Management** involves identifying, categorizing, and safeguarding valuable proprietary information. It includes creating trade secret policies, implementing security measures, and managing access to confidential information.
5. **Industrial Design Management:** Managing industrial designs entails protecting products' aesthetic and ornamental aspects.
6. **IP Portfolio Management:** This approach involves overseeing and strategically managing an organization's entire portfolio of IP assets. It includes assessing the value of each purchase, determining its relevance to business goals, and optimizing the portfolio for maximum value.

Depending on the size and structure of the organization, IP management may involve different roles and responsibilities:

1. **Chief Intellectual Property Officer (CIPO) or Chief IP Counsel:** In larger organizations, a dedicated CIPO or Chief IP Counsel might oversee all aspects of IP management. They lead IP strategy development, portfolio management, enforcement actions, and coordination with legal teams.
2. **Legal Department:** The legal team is crucial in IP management, handling patent, trademark, and copyright filings, and legal enforcement against infringements. Attorneys provide legal advice on IP matters, draft agreements, and represent the company in litigation.
3. **IP Manager or IP Coordinator:** This role focuses on day-to-day IP administration. IP managers handle tasks such as maintaining IP records, managing deadlines for renewals and filings, and coordinating with external IP counsel.
4. **R&D and Innovation Teams:** These teams generate new intellectual property through research and development efforts. They work closely with legal and IP professionals to identify patentable inventions, conduct patent searches, and draft patent applications.

Benefits of IP Management:

1. Share of IP and knowledge assets in the firm valuation is increasing and these assets deserve to have adequate attention.
2. IP management can be crucial for the survival of an organization in the emerging knowledge economy.
3. Value derived from IP may be undermined in the absence of proper IP management.
4. Sound IP management is the pre-requisite for the sustenance of IP ecosystem in the organization.
5. Value originating from IP of an organization may be captured by other organization if safeguards through good IP management are not put in place.
6. Superior IP management can benefit the customers, in particular and society, in general through improved product and services.

IP MANAGEMENT ACTIVITIES:

The IP management activities include

- Undertaking Ip Intelligence
 - Acquisition Of Ip
 - Decision On Ip Portfolio
 - Supporting Development of Ip
 - Commercialization Of Ip
 - Protecting Ip Management
 - Putting Ip Management in Place
 - Carrying Out Ip Audits
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- **UNDERTAKING IP INTELLIGENCE:** *Analyzing competitor activities and being well-informed of innovative new technologies is essential to a good Intellectual Property (IP) strategy.* Intellectual property (IP) intelligence and search refers to a process of gathering and analyzing IP-related data to provide businesses with relevant information on their industry and technologies. Conducting IP searches can also help the company benchmark its technology against existing ones.
 - **ACQUISITION OF IP:** Copyrights, trade- marks, and service marks may be obtained through creation, license or assignment and transfer. Registration of these IP types provides additional protections. Once obtained, intellectual property is like any other property; it can be sold, licensed, given away, or even forfeited.
 - **DECISION ON IP PORTFOLIO:** Intellectual Property Portfolio Management is the strategic administration of a company's intellectual assets, encompassing patents, trademarks, copyrights, and trade secrets. The combination of the intellectual property rights is termed portfolio. The company should decide the best combination where the benefit is more than the cost. It involves a systematic approach to acquiring, safeguarding, enforcing, and leveraging these intellectual property rights to bolster a company's competitive edge and overall worth.

- **SUPPORTING DEVELOPMENT OF IP:** The Intellectual Property (IP) system arises as an opportunity to turn those innovative ideas, creative designs and other intangible assets into valuable business assets, which can be transferred, licensed or used. The company should provide scope for research and development and provide resources to support development of intellectual property within the organization.
- **COMMERCIALIZATION OF IP:** IP commercialization transforms an idea or innovation into a product or service which is profitable in nature. The sale of such a product or service is the most prevalent practice used for monetization of creativity. IPRs play a very significant role in such circumstances. It prevents the illegal exploitation of protected features by the competitors.
For instance, when a medical company comes up with new medicine, its formula is a result of long-term investment and thus, the company should be able to yield the returns of the investment.
- **PROTECTING IP MANAGEMENT:** When something new and innovative is created, the inventor has the right to protect it. Inventors can register their creations as intellectual properties by filing for an IP that enables them to avail sole monopoly over their creations. As such, the right to protect IP entails the exclusive rights provided to the innovator or creator. IP rights denotes the legal rights provided to the creator and not the IP itself. It is necessary to understand the meaning of IP rights and be aware of the steps taken to protect them against infringement.
- **PUTTING IP MANAGEMENT IN PLACE:** The company should continuously involve is the way of managing intellectual property like; patent renewals, integration, and search.
- **CARRYING OUT IP AUDITS:** Conducting an audit of your existing intellectual property materials should be a part of your IP strategy development. Be sure to include utility patents, trademarks, copyrights, and design patents and the ownership and historical relationships related to these properties.

The scope of protection of the intellectual property should also be included in the intellectual property audit to determine whether there may be gaps in protection or risks in the development and intellectual property protection process. Finally, a formal exit strategy should be determined if problems arise at any stage of the development process.

5CS MODEL OF MANAGING IP

- **CARVING**
It is important to have a clear and **concise IP policy in place**. This policy should be accessible to all employees and outline the company's position on intellectual property. Secondly, all employees should be **made aware of the policy** and be trained on how to comply with it. Larger companies may want to outline the roles and responsibilities of managers and employees in managing and disseminating the company policy regarding

intellectual property used to ensure everyone follows the proper procedures to maintain the company's intellectual property rights.

- **CREATING**

Develop a pool of well informed and trained human resource, deploy sufficient facilities (hardware & software) and create and promote an enabling environment for generating, protecting and managing intellectual property for progress of science, technology and arts leading to growth of trade and industry.

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- **CAPTURING:** Of course, not all ideas are good ideas and even if there are good ideas, they may not be commercialized. If you implement a process to identify, capture and protect ideas, the business will preserve the ideas that can add value to the business.

The process should include a central person or team to whom employees can report innovations they make or solutions to problems they encounter that may be applied more broadly; have a standard form that can be used to explain what the innovation or solution is and how it works; and provide training on IP within the business to increase general awareness.

The most effective methods to stimulate and capture creativity in an organization are to provide a framework that recognizes IP in the business, establish a strong culture of support for creative thinking, provide mechanisms to recognize efforts by employees, allow freedom to explore new concepts, and facilitate regular discussions of IP issues.

To paraphrase a classic one-liner from a great movie: 'If you build it, they will come'. If you provide the structure and training, your employees will more than likely jump on board!

- **CONSERVING:**

Intellectual property law allows owners, inventors, and creators of intellectual property to protect themselves from unauthorized use of their property. The primary forms of intellectual property are patents (utility and design), trademarks, and copyrights.

Patents protect an inventor's creation, while trademark protection safeguards distinctive words, names, symbols, sounds, and colors used to distinguish one business's products and services from another.

Property that qualifies for copyright protection includes literary works such as books and computer programs; dramatic works and the accompanying words; pictorial, graphics, photographs, and sculptural works; motion pictures and audiovisual works; and sound

recordings and musical works, including music from plays and dramatic readings, and recordings on tapes, cassettes, and CDs.

RESEARCH AND DEVELOPMENT IN INDIA (CASES)

Research: IPR research is an emerging area and the research activities and results are having wider scope in terms of their impact on the people in the country. Intellectual Property Rights have gained a lot of recognition in the past few years in India mainly due to globalized economy, economic activity in India being on a healthy growth path and the efforts made by the Government to create enabling milieu by way of modernizing the IPR infrastructure as well as implementing various programmes for creation of awareness among the professionals as well as general public. Most research that is going on as of now are based on individual initiatives or sponsored research by corporation, which arguably have a very limited scope. The academic or research activities also require quality human resources to man the research positions as well as financial and technical resources in order to conduct any meaningful in depth, multidisciplinary and/or mature research in the field of Intellectual Property.

This gap has been recognized by the government and therefore, it is proposed to include research by the following institutions;

- Council of Scientific & Industrial Research (CSIR)
- Indian Council of Agricultural Research (ICAR)
- Indian Council of Medical Research (ICMR)
- Defence Research and Development Organisation (DRDO)
- Department of Atomic Energy (DAE)
- Department of Space (DoS)
- Department of Electronics (DoE)

INCENTIVES TO RESEARCH AND DEVELOPMENT:

The following incentives are provided for research and development related to intellectual properties in India

- 100% write off of revenue expenditure on R&D;
- Exemption from Excise Duty and Customs duty
- Exemption from Price Control of Drugs
- Support in the form of grants for industrial R&D projects through Technology Development and Innovation Programme (TDIP)
- Associations of industry with the national R&D programmes in area of atomic energy space and Defence research

Intellectual property in India:

- Recently, India ranks **42nd among 55 leading global economies on the International Intellectual Property (IP) Index 2023** released by the U.S. Chambers of

Commerce, according to which India is ripe to become a leader for emerging markets seeking to transform their economy through **IP-driven innovation**.

COMPANIES INVOLVED IN IP FOR AIML

- Tata Consultancy Services
- , Wipro
- Infosys
- HCL Technologies and
- IBM.

CHALLENGES IN INDIAN R&D

1. Indian R&D expenditure levels are lagging as compared to leading countries at the global platform.
2. There is an urgent need to prop up R&D expenditure and IP generation in the higher education sector.
3. There is need for better IP management so that IP assets can be put through the phase of commercialization.
4. There is a need on part of private sector to boost R&D expenditure.